

Stakeholder

Protocol > architecture · buy safer collection

Slogan

Label recoveries by disturbing the expert, not by deploying the learner.

Protocol > architecture; heed ACT's warning on fine teleop.

49→79%

HSR success

3×

compute win

−5% vs −80%

safer collect

+62% rel

vs BC

No new arch

BC pipeline fit

Talk arc — buy the idea, not the 2017 stack

Decision framing for Data IL programs

1. What you buy: safer/cheaper recovery labels without DAgger early catastrophes

2. Headline ROI: HSR 49→79 · Humanoid 3× · collect −5% vs −80%

3. Protocol > architecture: noise schedule + α — fits existing BC

4. Moats: MuJoCo parity + real human study + isotropic ablation honesty

5. Risks: α too aggressive · humans degrade · still iterative · coverage still needed

6. Decision: adopt recovery-augmentation recipe for teleop programs

7. 2017 HSR CNN $\neq$ product — buy the idea for 2026 stacks

8. Tattoo + close

What to buy

Demo protocol that buys robustness without catastrophic early rollouts

Safer collection

Supervisor stays near manifold; Humanoid collect reward only -5% vs clean expert (DAgger -80%).

Compute win

Up to 3× wall-clock vs naive DAgger retrain-every-demo on Humanoid-class control.

Human study lift

Toyota HSR clutter: 49% $\rightarrow$ 79% with modest $\alpha=3$; $\sim +62\%$ relative over BC.

Pipeline fit

No new architecture — only collection protocol + light iterative noise/policy refits on BC.

Headline ROI — tattoo numbers

Quote these in every stakeholder briefing

49% → 79%

HSR BC → DART($\alpha=3$)

72%

DART($\alpha=6$) — still > BC
but α too high for humans

+62% rel

Average performance
increase over BC

3x

Humanoid wall-clock
vs DAgger

-5%

DART supervisor reward
during collection

-80%

DAgger policies' collect
reward vs supervisor

α inv-U

Moderate noise wins
for human teleop

Protocol > architecture

The durable artifact is a demo-collection recipe

1 Noise model

Gaussian on continuous U
(or ϵ -greedy discrete)

2 Fit Σ

MLE outer-product of
(learner – expert) actions

3 α -shrink

Scale so expected error $\approx \alpha$
(prior on final learner error)

4 Collect

Supervisor teleops under
 ψ^α — demos contain recoveries

5 Aggregate BC

Standard supervised IL on
noisy + clean demos

6 Iterate K

HSR used $K=2$; MuJoCo few
updates — not DAgger loops

Why not just buy DAgger?

Same shift objective — opposite sampling locus

	DAgger	DART
Who acts	Mixture / learner	Supervisor + action noise
Who labels	Supervisor on learner states	Noisy expert demos
Retrain cadence	Every / frequent demos	Infrequent ψ + final BC
Danger early?	Often yes	Near expert funnel
Human load	High (fix under failures)	Lower (small disturbances)
Collect reward	~–80% Humanoid	~–5% Humanoid

Moats & evidence

What makes the buy defensible

MuJoCo parity

DART $\approx$ DAgger final reward on
Walker/Hopper/HalfCheetah/Humanoid

Compute + safety

3 $\times$ wall-clock $\cdot$ collect -5% vs -80% — ops metrics,
not just SR

Isotropic ablation

$\Sigma=I$ fails — optimized Σ^α is the product, not “add
noise” folklore

Real humans

4 supervisors, within-subject HSR clutter — $\alpha=3$
beats $\alpha=6$ (honest)

Theory spine

Lemma 4.1 KL bound $\cdot$ Prop 4.1 noisy expert strictly
closer when error > 0

Open artifact

Code + BAIR blog — recipe is reproducible in spirit

Risks / costs

What can go wrong if you buy carelessly

α too aggressive

Humans degrade; HSR $\alpha=6$ (72%) < $\alpha=3$ (79%).
Inverted-U is real.

Noise model class

Need Gaussian / ϵ -greedy + held-out demos for Σ .
Not free.

Still iterative

Not one-shot BC; $K \geq 2$ rounds and covariance estimation overhead.

Coverage still needed

Prop 4.1: if learner fits expert perfectly, noise adds little.

2017 stack $\neq$ product

CNN pushing + open-loop grasp far from modern VLA —
idea travels, code doesn't.

Supervisor unaffected?

Theory assumes noise doesn't change expert intent;
humans may violate.

Decision takeaway

For Data-oriented IL programs

ADOPT

Treat DART as a reference recovery-augmentation recipe: inject optimized action noise during teleop so recovery is labeled in-distribution.

Buy safer collection + 49→79 + 3× compute idea — not the 2017 HSR codebase.

Do now

Add α -scheduled action-noise
card to teleop SOPs
Log collect-time reward

Pilot

Hopper/ALOHA-lite:
BC vs isotropic vs DART(α)
before robot-days

Don't

Blind isotropic on fine
bimanual demos
Claim VLA “is DART”

Action noise $\neq$ state/obs noise

Buy the right knob

DART (buy this)

Noise on supervisor control u
→ dynamics widen state dist
→ recovery labels near boundary
Optimized Σ^α , not $\Sigma=I$
 α inverted-U for humans

Not the same

Observation dropout
Visual domain rand
Pixel augment
State jitter on sensors
These attack perception shift

Also complementary

Action chunking (ACT/DP)
attacks effective horizon
DART attacks data dist
Use both; task-gate noise
(ACT warning for fine contact)

Evidence board — HSR 49 / 79 / 72

Fig. 3 · human study you can put in a memo



Figure 3: Left: Experimental setup for the grasping in clutter task. A Toyota HSR robot uses a head-mounted RGBD camera and its arm to push obstacle objects out of the way to reach the goal object, a mustard bottle. The robot's policy for pushing objects away uses a CNN trained on images taken from the robot's Primesense camera, an example image from the robot's view point is shown in the orange box. Right: the Success Rate for Behavior Cloning, DART($\alpha = 3$) and DART($\alpha = 6$). DART($\alpha = 3$) achieves the largest success rate.

the number of times the policy is updated. DAgger-B illustrates that this can significantly deteriorate performance. Lastly, naively applying isotropic noise does not perform well, and leads to unsafe policies during execution, which suggests the need for optimizing the level of noise.

5.2 Robotic Grasping in Clutter

We evaluate DART with human supervisors in a grasping in clutter task on a Toyota HSR robot. We consider a task inspired by the Amazon Picking Challenge [2]. The goal of the task is for the robot to retrieve a goal object from a cupboard. The task is challenging because the goal object is occluded by obstacle objects and the robot must reason about how to clear a path based on observations taken from an eye-in-hand perspective. The objects are 6 common household food items, which consist of boxes and bottles with varying textures and mass distributions. The target object is fixed to always be a mustard bottle. The robot, task and image viewpoint are shown in Fig. 3. See supplement material for additional information on the task.

We use 4 supervisors who have robotics experience but not specifically in the field of Imitation Learning and compare Behavior Cloning and DART. When performing the study, we first collect $N = 10$ demonstrations with Behavior Cloning (i.e. no noise) and then in a counter-balanced ordering collect $N = 30$ more demonstrations with each technique. Our experiment was within-subject, have every supervisor perform all three methods.

We only updated the noise parameter after the first 10 demonstrations (i.e. $K = 2$). The final robot policy was then trained on the total of 40 demonstrations. We consider two different choices of α : $\alpha = 3\text{Tr}(\Sigma_1)$ and $\alpha = 3T\text{tr}(\Sigma_1)$. These choices correspond to the intuition that with small datasets in high dimensional image space the robot will have significant shift from the current loss on the supervisor's distribution. Through the rest of the paper, we will write our choices as $\alpha = 3$ and $\alpha = 6$ for brevity in notation.

During policy evaluation, we measured success as 1) the robot is able to identify the goal object and 2) there being a clear path between its gripper and the object. Once these conditions are identified, as detailed in the supplement material, an open-loop motion plan is generated to execute a grasp around the target object.

In Fig. 3, we report the average performance of the three techniques across DART 20 times on different sampled initial states. DART with $\alpha = 3$ performs the best out of the three techniques, with a 79% success rate over traditional Behavior Cloning's 49% success rate. Interestingly, DART with $\alpha = 6$ performs better than Behavior Cloning, but only has a 72% success rate. This may suggest this level of noise was potentially too high for the human supervisor.

6 Conclusion

This paper considers injecting noise into the supervisor's demonstrations to minimize covariate shift. Our algorithm, DART, provides the robot with corrective examples at the boundary of the supervisor's policy. By being concentrated around the supervisor's policy, it collects demonstrations without visiting highly sub-optimal states and is easier for human supervisors tele-operating. We demonstrate it achieves parity with on-policy methods in simulated MuJoCo domains and is significantly better than traditional Behavior Cloning on a real robot. For code and more information see <http://berkeleyautomation.github.io/DART/>

Evidence board — MuJoCo 3× / −5% / −80%

Fig. 2 · compute + collect-safety story

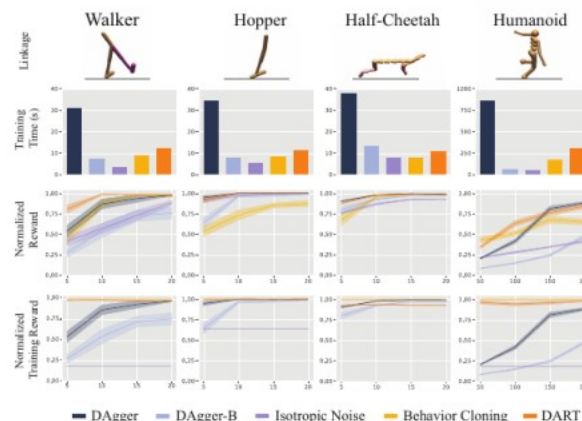


Figure 2: Top: The four different locomotive domains in MuJoCo we evaluated DART on: Walker, Hopper, Half-Cheetah and Humanoid. Top Middle: The time, in seconds, to achieve the performance level reported below. DART achieves similar performance to DAgger in all 4 domains, but requires significantly less computation because it doesn't require retraining the current robot policy after each demonstration. DAgger-B reduces the computation required by less frequently training the robot, but suffers significantly in performance in domains like Humanoid. Bottom Middle: The learning curve with respect to reward obtained during training with each algorithm plotted across the number of demonstrations. Bottom: The reward obtained during collection of demonstrations for learning. DART receives near the supervisor's reward at all iterations whereas DAgger can be substantially worse in the beginning.

policies in Tensorflow. At each iteration we collected one demonstration. In order to make the task challenging for Imitation Learning, we used the same technique as in [6], which is to sub-sample 50 state and control pairs from the demonstrated trajectories, making the learners receive less data at each iteration.

We compare the following five techniques: Behavior Cloning, where data is collected from the supervisor's distribution without noise injected; DAgger, which is an on-policy method that stochastically samples from the robot's distribution and then fully retrain the robot's policy after every demonstration is collected; DAgger-B, which is DAgger but the retraining is done intermittently to reduce computation time; Isotropic-Gaussian which injects non-optimized isotropic noise; and DART. See supplement material for how the hyper-parameters for each method are set.

We used the following MuJoCo environments: Walker, Hopper, Humanoid, and Half-Cheetah. The size of the state and control space for each task is $\{|\mathcal{X}| = 17, |\mathcal{U}| = 6\}$, $\{|\mathcal{X}| = 11, |\mathcal{U}| = 3\}$, $\{|\mathcal{X}| = 376, |\mathcal{U}| = 17\}$, $\{|\mathcal{X}| = 117, |\mathcal{U}| = 6\}$, respectively. All experiments were run on a MacBook Pro with an 2.5 GHz Intel Core i7 CPU. We measured the cumulative reward of each learned policy by rolling it for 500 timesteps, the total computation time for each method and the cumulative reward obtained during learning.

Fig. 2 shows the results. In all domains, DART achieves parity with DAgger, whereas Behavior Cloning and DAgger-B are below this performance level in Walker and Humanoid. For Humanoid, DART is 3x faster in computation time and during training only decreases the supervisor's cumulative reward by 5%, whereas DAgger executes policies that have over 80% less cumulative reward than the supervisor. The reason for this is that DAgger requires constantly updating the current robot policy and forces the robot into sub-optimal states. While, one way to reduce this computation is to decrease

Tattoo card

HSR 49% / 79% / 72% · +62% rel · Humanoid 3× wall-clock · collect −5% (DART) vs −80% (DAgger) · α inverted-U

49%

BC

79%

DART-3

72%

DART-6

+62%

rel lift

3×

wall

−5%

DART collect

−80%

DAgger collect

α inv-U

3 > 6

$\Sigma^{\alpha} \gg I$

opt > iso

Spine: Safer/cheaper than DAgger · HSR 49→79% · Humanoid 3× · collect −5% vs −80% · action≠state noise · $\Sigma^{\alpha} \gg$ isotropic · α inverted-U · complemen...

Close — Stakeholder

Four asks

1. Buy the recovery-augmentation protocol, not the 2017 HSR CNN.
2. Insist on optimized $\Sigma^\alpha + \alpha$ calibration — isotropic is not DART.
3. Track collect-time reward and wall-clock alongside success rate.
4. Task-gate noise for fine contact (ACT warning); pair with chunking.